AJ_24038155_dialect_v1.2

Full integrated Cambridge-Style Pseudocode specification

Integrated specification: This document incorporates the complete v1.0 dialect and the approved v1.1 and v1.2 language additions into a single normative specification.

1. Language Overview

The AJ_24038155 Dialect is a structured pseudocode language intended for:

- representing algorithms generated from natural-language tasks;
- deterministic parsing into an Abstract Syntax Tree;
- conversion into a Control Flow Graph;
- visual generation of flow diagrams;
- deterministic translation into Python;
- safe behavioural evaluation.

The language uses explicit block terminators and does not rely on indentation for program structure.

Keywords are case-insensitive during parsing but are presented in uppercase as the canonical format.

Identifiers remain case-sensitive.

2. Lexical Rules

2.1 Identifiers

An identifier must begin with a letter or underscore.

```
identifier ::= letter , { letter | digit | "_" } ;
```

Examples:

total

item_count

_grid

value2

Invalid examples:

2value

item-count

total value

2.2 Integer Literals

```
integer_literal ::= [ "-" ] , digit , { digit } ;
```

Examples:

0

12

-5

1000

2.3 Float Literals

```
float_literal ::= [ "-" ] , digit , { digit } , "." , digit , { digit } ;
```

Examples:

0.5

3.14

-10.25

2.4 String Literals

Strings use the STRING constructor with quoted contents.

```
string_literal ::= "STRING" , "(" , quoted_string , ")" ;
```

Examples:

```
STRING("Hello")
STRING("Enter a number")
STRING("The result is {result}")
```

Escape sequences may be used:

\"

\\

\n

\t

Example:

```
STRING("He said \"Hello\"")
```

2.5 Character Literals

Characters use the CHAR constructor and must contain exactly one character or one valid escape sequence.

```
char_literal ::= "CHAR" , "(" , quoted_character , ")" ;
```

Examples:

```
CHAR("a")
CHAR("7")
CHAR("\n")
```

2.6 Boolean Literals

```
boolean_literal ::= "BOOLEAN" , "(" , ( "TRUE" | "FALSE" ) , ")" ;
```

Canonical examples:

```
BOOLEAN(TRUE)
BOOLEAN(FALSE)
```

2.7 Native Null Literal

NONE is the native null literal in v1.1.

```
none_literal ::= "NONE" ;
```

Keywords are case-insensitive during parsing, therefore NONE, None and none denote the same literal.

Canonical output uses uppercase NONE.

Python translation:

```
NONE
```

translates deterministically to:

None

NONE is a value. It is not an identifier and is not a function call.

The historical compatibility form CALL NULL() remains accepted for existing v1.0 material. New canonical v1.1 pseudocode should use NONE.

2.8 Comments

Single-line comments begin with COMMENT.

```
comment ::= "COMMENT" , comment_text , newline ;
```

Example:

```
COMMENT Calculate the running total
```

Comments are ignored during AST construction but may be stored as metadata.

2.9 Whitespace and Newlines

Spaces and indentation are not semantically significant except inside string literals.

Each statement must end with a newline.

Block structure is determined by explicit keywords such as:

```
ENDIF
ENDWHILE
ENDFOR
ENDFUNCTION
```

3. Operator Rules

3.1 Assignment

Assignment uses <-.

```
variable <- value
```

Example:

```
total <- 0
name <- STRING("Alan")
```

3.2 Arithmetic Operators

+

-

*

/

```
INTEGER DIVISION
MODULUS
EXP
```

Examples:

```
sum <- a + b
difference <- a - b
```

```
product <- a * b
quotient <- a / b
whole <- a INTEGER DIVISION b
remainder <- a MODULUS b
power <- x EXP y
```

3.3 Comparison Operators

==

!=

<

>

<=

>=

Examples:

x == y

x != y

x<y

x >= y

3.4 Boolean Operators

```
NOT

AND
OR
XOR
```

Examples:

x > 0 AND x < 10

NOT finished

left OR right

flag1 XOR flag2

3.5 Operator Precedence

Operators are evaluated in the following order, from highest to lowest:

1. Parentheses

2. Function and collection access

3. EXP

4. Unary minus

5. *, /, INTEGER DIVISION, MODULUS

6. +, -

7. ==, !=, <, >, <=, >=

8. NOT

9. AND

10. XOR

11. OR

Parentheses should be used where ambiguity may occur.

4. Data Types and Literals

4.1 Supported Primitive Types

INTEGER
FLOAT
STRING
CHAR
BOOLEAN

4.2 Integer Literal

INTEGER(5)

INTEGER(-20)

Formal rule:

typed_integer ::= "INTEGER" , "(" , integer_literal , ")" ;

4.3 Float Literal

FLOAT(3.14)
FLOAT(-0.5)

Formal rule:

typed_float ::= "FLOAT" , "(" , float_literal , ")" ;

4.4 Tuple Literal

TUPLE(a, b)
TUPLE(INTEGER(1), STRING("A"))

Formal rule:

tuple_literal ::=

"TUPLE" ,

"(" ,

[expression , { "," , expression }] ,

")" ;

4.5 Homogeneous Typed Lists

A homogeneous list contains values of one declared type.

Examples:

LIST OF INTEGER [1, 2, 3]
LIST OF STRING [STRING("A"), STRING("B")]
LIST OF BOOLEAN [BOOLEAN(TRUE), BOOLEAN(FALSE)]

Formal rule:

typed_list_literal ::=

"LIST OF" ,

type_name ,

"[" ,

[expression , { "," , expression }] ,

"]" ;

All values must be compatible with the declared type.

4.6 Mixed Lists

Mixed lists use the LIST keyword.

Example:

```
LIST [INTEGER(5), STRING("hello"), CHAR("a"), BOOLEAN(TRUE)]
```

Formal rule:

```
mixed_list_literal ::=
```

"LIST" ,

"[" ,

[expression , { "," , expression }] ,

"]" ;

4.7 Empty Values

Supported empty values are:

```
EMPTY LIST
EMPTY DICTIONARY
EMPTY STRING
EMPTY SET
EMPTY TUPLE
```

Formal rule:

```
empty_literal ::=
```

"EMPTY LIST"

```
| "EMPTY DICTIONARY"
| "EMPTY STRING"
| "EMPTY SET"
| "EMPTY TUPLE" ;
```

4.8 Native Null Literal

NONE represents the absence of a value.

Canonical syntax:

```
NONE
```

Formal rule:

```
none_literal ::= "NONE" ;
```

Python translation:

NONE -> None

5. Program Structure

5.1 Program

A program contains one or more declarations or statements.

```
program ::= { declaration | statement | comment } ;
```

A program should normally contain at least one function when it is intended for automated testing.

5.2 Function Definition

Canonical syntax:

```
FUNCTION name ACCEPTS(parameter1, parameter2)
```

statements

```
ENDFUNCTION
```

Example:

```
FUNCTION calculate_total ACCEPTS(numbers)
total <- 0

FOR EACH number IN numbers DO
total <- total + number
ENDFOR

RETURN total
ENDFUNCTION
```

Formal rule:

```
function_definition ::=
"FUNCTION" ,
identifier ,
"ACCEPTS" ,
"(" ,
[ parameter_list ] ,
")" ,
newline ,
function_body ,
"ENDFUNCTION" ,
newline ;

function_body ::= { function_body_item } ;

function_body_item ::=
statement
| function_definition
| comment ;
```

5.2.1 Nested/local function declarations

A FUNCTION ... ENDFUNCTION declaration may appear inside the body of another function. A function declared in this way is a nested function, also called a local function.

Nested functions use the existing FUNCTION syntax and may themselves contain nested function declarations. The nested function name belongs to the lexical scope of the enclosing function and must not be promoted to module scope.

```
FUNCTION outer ACCEPTS(value)
   FUNCTION inner ACCEPTS(item)
      RETURN item
   ENDFUNCTION

   RETURN CALL inner(value)
ENDFUNCTION
```

A nested function may call itself recursively by name and may read values visible in an enclosing lexical scope. Assignment to a simple identifier inside the nested function follows the nested function local-binding rule; v1.2 introduces no NONLOCAL-style rebinding construct.

v1.2 adds declaration nesting only. It does not introduce callbacks, first-class function values, anonymous functions, or function-reference expressions.

Python translation preserves lexical nesting. A nested AJ function compiles to a nested Python def at the corresponding indentation; the compiler must not hoist it merely to simplify code generation.

5.3 Parameter List

parameter_list ::=

parameter ,

{ "," , parameter } ;

A parameter may optionally declare a type.

Examples:

```
FUNCTION process ACCEPTS(value)
FUNCTION process ACCEPTS(value AS INTEGER)
```

Formal rule:

parameter ::=
identifier ,

["AS" , type_name] ;

5.4 Return Statement

RETURN expression

or:

RETURN

Formal rule:

return_statement ::=

"RETURN" ,

[expression] ,

newline ;

A value-less return is permitted.

5.5 Procedure and Function Calls

A call may be used as an expression:

result <- CALL calculate_total(numbers)

or as a standalone statement:

CALL display_result(result)

Formal rules:

call_expression ::=

"CALL" ,

identifier ,

"(" ,

[argument_list] ,

")" ;

call_statement ::=

call_expression ,

newline ;

argument_list ::=
expression ,

{ "," , expression } ;

6. Input and Output

6.1 Input

Input must contain a prompt expression.

```
age <- INPUT(STRING("Enter your age"))
```

Formal rule:

input_expression ::=

"INPUT" ,

"(" ,

expression ,

")" ;

6.2 Output

OUTPUT expression

Examples:

```
OUTPUT result
OUTPUT STRING("The value is {value}")
```

Formal rule:

output_statement ::=

"OUTPUT" ,

expression ,
newline ;

String placeholders use braces:

```
STRING("The result is {result}")
```

The referenced identifier must exist in the current scope.

7. Assignment Rules

7.1 Standard Assignment

identifier <- expression

Example:

```
count <- 0
```

7.2 Collection Element Assignment

```
list[index] <- value
dictionary[key] <- value
```

Examples:

```
numbers[0] <- 10
visited[TUPLE(nr, nc)] <- TUPLE(r, c)
```

Formal rule:

```
assignment_statement ::=
```

assignable ,

"<-" ,

```
expression ,
newline ;

assignable ::=
```

identifier

```
| indexed_access ;
```

8. Expressions

8.1 Expression Grammar

```
expression ::= or_expression ;

or_expression ::=
```

xor_expression ,

{ "OR" , xor_expression } ;

```
xor_expression ::=
```

and_expression ,

{ "XOR" , and_expression } ;

```
and_expression ::=
```

not_expression ,

{ "AND" , not_expression } ;

```
not_expression ::=
```

"NOT" , not_expression

```
| comparison_expression ;

comparison_expression ::=
```

additive_expression ,

[comparison_operator , additive_expression] ;

```
additive_expression ::=
```

multiplicative_expression ,

{ ("+" | "-") , multiplicative_expression } ;

multiplicative_expression ::=

power_expression ,

{

(

"*"

| "/"
| "INTEGER DIVISION"
| "MODULUS"

),

power_expression

};

power_expression ::=

unary_expression ,

["EXP" , power_expression] ;

unary_expression ::=

"-" , unary_expression

| primary_expression ;

primary_expression ::=

literal

| identifier
| indexed_access
| member_call_expression
| call_expression
| input_expression
| length_expression
| join_expression
| dictionary_get_expression
| random_expression
| "(" , expression , ")" ;

8.2 Comparison Operator

comparison_operator ::=

"=="

| "!="
| "<"
| ">"
| "<="
| ">=" ;

9. Conditional Rules

9.1 IF Statement

IF condition THEN

statements

ENDIF

Formal rule:

if_statement ::=

"IF" ,

expression ,

"THEN" ,

newline ,

{ statement | comment } ,

{ else_if_clause } ,

[else_clause] ,

"ENDIF" ,

newline ;

9.2 ELSE IF Clause

ELSE IF condition THEN

statements

Formal rule:

else_if_clause ::=

"ELSE IF" ,

expression ,

"THEN" ,

newline ,

{ statement | comment } ;

9.3 ELSE Clause

ELSE

statements

Formal rule:

else_clause ::=

"ELSE" ,

newline ,

{ statement | comment } ;

9.4 Full Example

```
IF score >= 70 THEN
grade <- STRING("Distinction")
ELSE IF score >= 60 THEN
grade <- STRING("Merit")
ELSE
grade <- STRING("Pass")
ENDIF
```

10. Loop Rules

10.1 WHILE Loop

WHILE condition DO

statements

ENDWHILE

Formal rule:

while_statement ::=

"WHILE" ,

expression ,

"DO" ,

newline ,

{ statement | comment } ,

"ENDWHILE" ,

newline ;

10.2 Range Loop

Range loops are end-exclusive.

Therefore:

FOR i <- IN RANGE 0 TO n DO

iterates through:

0, 1, 2, ..., n - 1

Canonical syntax:

FOR identifier <- IN RANGE start TO end DO

statements

ENDFOR

Optional step:

FOR i <- IN RANGE 10 TO 0 STEP -1 DO

statements

ENDFOR

Formal rule:

range_for_statement ::=

"FOR" ,

identifier ,

"<-" ,

"IN RANGE" ,

expression ,

"TO" ,

expression ,

["STEP" , expression] ,

"DO" ,

newline ,

{ statement | comment } ,

"ENDFOR" ,

newline ;

A step value must not equal zero.

10.3 FOR EACH Loop

FOR EACH item IN collection DO

statements

ENDFOR

Formal rule:

for_each_statement ::=

"FOR EACH" ,

identifier ,

"IN" ,

expression ,

"DO" ,

newline ,

{ statement | comment } ,

"ENDFOR" ,

newline ;

10.4 BREAK

BREAK

Formal rule:

break_statement ::=

"BREAK" ,

newline ;

BREAK is valid only inside:

WHILE
FOR
FOR EACH

10.5 CONTINUE

The dialect includes an optional CONTINUE statement.

CONTINUE

Formal rule:

continue_statement ::=

"CONTINUE" ,

newline ;

CONTINUE is valid only inside a loop.

11. List and Collection Rules

11.1 Indexing

collection[index]

Examples:

numbers[0]

grid[row][column]

Formal rule:

indexed_access ::=

primary_identifier ,

{ "[" , expression , "]" } ;

primary_identifier ::= identifier ;

Indexes begin at zero.

11.2 Length

LENGTH OF collection

Formal rule:

length_expression ::=

"LENGTH OF" ,

expression ;

11.3 Append

APPEND value TO collection

Formal rule:

append_statement ::=

"APPEND" ,

expression ,

"TO" ,

expression ,
newline ;

11.4 Insert

INSERT value INTO collection AT index

Formal rule:

insert_statement ::=

"INSERT" ,

expression ,

"INTO" ,

expression ,

"AT" ,

expression ,
newline ;

11.5 Remove First Item

REMOVE FIRST ITEM FROM collection

Formal rule:

remove_first_statement ::=

"REMOVE FIRST ITEM FROM" ,

expression ,
newline ;

11.6 Remove Last Item

REMOVE LAST ITEM FROM collection

Formal rule:

remove_last_statement ::=

"REMOVE LAST ITEM FROM" ,

expression ,
newline ;

11.7 Remove Specific Item

REMOVE value FROM collection

Formal rule:

remove_value_statement ::=

"REMOVE" ,

expression ,

"FROM" ,

expression ,
newline ;

11.8 Join

JOIN(collection, separator)

Example:

result <- JOIN(characters, EMPTY STRING)

Formal rule:

join_expression ::=

"JOIN" ,

"(" ,

expression ,

"," ,

expression ,

")" ;

11.9 Two-Dimensional List Creation

Canonical form:

```
grid <- EMPTY LIST
r <- 0

WHILE r < rows DO
row <- EMPTY LIST
c <- 0

WHILE c < cols DO
APPEND fill_value TO row
c <- c + 1
ENDWHILE

APPEND row TO grid
r <- r + 1
ENDWHILE
```

No special two-dimensional list syntax is required. Nested lists are constructed using ordinary list

operations.

12. Set Rules

12.1 Set Literal

```
SET [value1, value2, value3]
```

Formal rule:

```
set_literal ::=
```

"SET" ,

"[" ,

[expression , { "," , expression }] ,

"]" ;

12.2 Set Operations

```
UNION(set1, set2)
INTERSECTION(set1, set2)
DIFFERENCE(set1, set2)
```

Formal rules:

```
set_operation ::=
```

"UNION" , "(" , expression , "," , expression , ")"

```
| "INTERSECTION" , "(" , expression , "," , expression , ")"
| "DIFFERENCE" , "(" , expression , "," , expression , ")" ;
```

12.3 Membership

value IN collection

value NOT IN collection

These are treated as comparison expressions.

13. Dictionary Rules

13.1 Dictionary Creation

```
visited <- EMPTY DICTIONARY
```

13.2 Dictionary Assignment

```
dictionary[key] <- value
```

Example:

```
visited[TUPLE(nr, nc)] <- TUPLE(r, c)
```

13.3 Dictionary Retrieval

Canonical v1.1 syntax:

```
value <- dictionary.GET(key)
```

Example:

```
node <- visited.GET(node)
```

Dictionary retrieval uses the general member-method invocation notation.

13.4 Dictionary Retrieval With Default

Optional syntax:

```
value <- dictionary.GET(key, default_value)
```

When resolved as the dictionary intrinsic, GET accepts exactly one or two arguments:

```
dictionary.GET(key)
dictionary.GET(key, default_value)
```

Without a default value, a missing key produces a runtime error.

13.4.1 Semantic Resolution

GET is not globally interpreted as dictionary retrieval by spelling alone.

The source form receiver.GET(...) is first parsed as an ordinary member-method invocation. It resolves to the dictionary intrinsic only when static evidence establishes that receiver is dictionary-backed, for example:

```
frequency <- EMPTY DICTIONARY
count <- frequency.GET(character, 0)
```

or:

```
FUNCTION lookup ACCEPTS(values AS DICTIONARY, key)
RETURN values.GET(key)
ENDFUNCTION
```

or when a class property is declared AS DICTIONARY.

If the receiver is not dictionary-backed, GET remains an ordinary user-defined method name.

Python translation:

```
dictionary.GET(key) -> dictionary.get(key)
dictionary.GET(key, default_value) -> dictionary.get(key, default_value)
```

13.4.2 Backwards Compatibility With v1.0

The v1.0 spellings dictionary~GET(key) and dictionary~GET(key, default_value) remain accepted for backwards compatibility.
Under v1.1 compatibility normalisation they canonicalise to dictionary.GET(key) and dictionary.GET(key, default_value).

The ~GET spelling is accepted but non-canonical in v1.1.

13.5 Dictionary Membership

key IN dictionary

key NOT IN dictionary

14. Random Rules

14.1 Random Integer

RANDOM INT(minimum, maximum)

Both endpoints are inclusive.

Formal rule:

random_integer_expression ::=

"RANDOM INT" ,

"(" ,

expression ,

"," ,

expression ,

")" ;

14.2 Random Float

RANDOM FLOAT(minimum, maximum)

Formal rule:

random_float_expression ::=

"RANDOM FLOAT" ,

"(" ,

expression ,

"," ,

expression ,

")" ;

14.3 Random Boolean

RANDOM BOOLEAN()

Formal rule:

random_boolean_expression ::=

"RANDOM BOOLEAN" ,

"(" ,

")" ;

14.4 Combined Rule

random_expression ::=

random_integer_expression

| random_float_expression
| random_boolean_expression ;

15. Error Handling

15.1 TRY-CATCH

TRY

statements

CATCH error

statements

ENDTRY

The error identifier is optional.

TRY

statements

CATCH

statements

ENDTRY

Formal rule:

try_statement ::=

"TRY" ,

newline ,

{ statement | comment } ,

"CATCH" ,

[identifier] ,

newline ,

{ statement | comment } ,

"ENDTRY" ,

newline ;

15.2 RAISE

Optional explicit error generation:

RAISE expression

Example:

```
RAISE STRING("Invalid input")
```

Formal rule:

```
raise_statement ::=

"RAISE" ,

    expression ,
    newline ;
```

16. Class Rules

Class support is included but may be excluded from the first experimental subset if the 50 algorithms do

not require it.

16.1 Class Definition

```
CLASS ClassName

declarations

ENDCLASS
```

Formal rule:

```
class_definition ::=

"CLASS" ,

    identifier ,

newline ,

{ constructor_definition | method_definition | property_declaration | comment } ,

"ENDCLASS" ,

    newline ;
```

16.2 Constructor

The dialect uses THIS instead of Python's self.

```
CONSTRUCTOR ACCEPTS(x, y)
PROPERTY THIS.x <- x
PROPERTY THIS.y <- y
ENDCONSTRUCTOR
```

Formal rule:

```
constructor_definition ::=

"CONSTRUCTOR" ,

"ACCEPTS" ,

"(" ,

    [ parameter_list ] ,

")" ,

newline ,

{ statement | property_assignment | comment } ,

"ENDCONSTRUCTOR" ,
```

newline ;

16.3 Property Declaration

PROPERTY name

or:

PROPERTY name AS INTEGER

Formal rule:

property_declaration ::=

"PROPERTY" ,

identifier ,

["AS" , type_name] ,

newline ;

16.4 Property Assignment

PROPERTY THIS.name <- value

Formal rule:

property_assignment ::=

"PROPERTY" ,

"THIS" ,

"." ,

identifier ,

"<-" ,

expression ,
newline ;

16.5 Method Definition

METHOD name ACCEPTS(parameters)

statements

ENDMETHOD

Formal rule:

method_definition ::=

"METHOD" ,

identifier ,

"ACCEPTS" ,

"(" ,

[parameter_list] ,

")" ,

newline ,

{ statement | comment } ,

"ENDMETHOD" ,

newline ;

16.6 Property Access

THIS.property_name

object.property_name

Formal rule:

property_access ::=
identifier ,

"." ,

identifier ;

THIS is treated as a reserved contextual identifier inside classes.

16.7 Member-Method Invocation

v1.1 explicitly distinguishes property access from invocation by the presence of parentheses.

Canonical syntax:

object.method(arguments)

Examples:

stack.pop()

drone.move(nextMove)

registry.GET(name)

Formal rule:

member_call_expression ::=

primary_expression ,

"." ,

identifier ,

"(" ,

[argument_list] ,

")" ;

The method identifier remains case-sensitive.

Parentheses distinguish a method invocation from property access:

x.item COMMENT property access
x.item() COMMENT zero-argument method invocation

A member call may be used as an expression or as a standalone statement.

17. Type Names

type_name ::=

"INTEGER"

```
    | "FLOAT"
    | "STRING"
    | "CHAR"
    | "BOOLEAN"
    | "LIST"
    | "DICTIONARY"
    | "SET"
    | "TUPLE"
    | identifier ;
```

A user-defined class name may be used as a type.

18. Statement Grammar

```
    statement ::=

assignment_statement

    | output_statement
    | call_statement
    | append_statement
    | insert_statement
    | remove_first_statement
    | remove_last_statement
    | remove_value_statement
    | if_statement
    | while_statement
    | range_for_statement
    | for_each_statement
    | return_statement
    | break_statement
    | continue_statement
    | try_statement
    | raise_statement ;
```

19. Declaration Grammar

```
    declaration ::=

function_definition

    | class_definition ;
```

20. Literal Grammar

```
    literal ::=

integer_literal

    | float_literal
    | typed_integer
    | typed_float
    | string_literal
    | char_literal
    | boolean_literal
    | none_literal
    | tuple_literal
    | typed_list_literal
    | mixed_list_literal
    | set_literal
    | empty_literal ;
```

21. Complete High-Level Grammar

```
    program ::=
    { declaration | statement | comment } ;

    declaration ::=
```

```
    function_definition
    | class_definition ;

    function_body ::= { function_body_item } ;

    function_body_item ::=
    statement
    | function_definition
    | comment ;

    statement ::=
    assignment_statement
    | output_statement
    | call_statement
    | append_statement
    | insert_statement
    | remove_first_statement
    | remove_last_statement
    | remove_value_statement
    | if_statement
    | while_statement
    | range_for_statement
    | for_each_statement
    | return_statement
    | break_statement
    | continue_statement
    | try_statement
    | raise_statement ;

    expression ::=

or_expression ;

    or_expression ::=

xor_expression ,

{ "OR" , xor_expression } ;

    xor_expression ::=

and_expression ,

{ "XOR" , and_expression } ;

    and_expression ::=

not_expression ,

{ "AND" , not_expression } ;

    not_expression ::=

"NOT" , not_expression

    | comparison_expression ;

    comparison_expression ::=

additive_expression ,

[

    (

"=="

    | "!="
    | "<"
    | ">"
    | "<="
    | ">="
```

```
    | "IN"
    | "NOT IN"

),

additive_expression

];

    additive_expression ::=

multiplicative_expression ,

{ ( "+" | "-" ) , multiplicative_expression } ;

    multiplicative_expression ::=

power_expression ,

{

    (

"*"

    | "/"
    | "INTEGER DIVISION"
    | "MODULUS"

),

power_expression

};

    power_expression ::=

unary_expression ,

[ "EXP" , power_expression ] ;

    unary_expression ::=

"-" , unary_expression

    | primary_expression ;

    primary_expression ::=

literal

    | identifier
    | indexed_access
    | property_access
    | member_call_expression
    | call_expression
    | input_expression
    | length_expression
    | join_expression
    | dictionary_get_expression
    | random_expression
    | set_operation
    | "(" , expression , ")" ;

    member_call_expression ::=

primary_expression ,

"." ,

    identifier ,
```

"(" ,

[argument_list] ,

")" ;

none_literal ::= "NONE" ;

22. Static Semantic Rules

The parser may produce a syntactically valid AST that still violates semantic rules. The following checks must therefore be performed after parsing.

22.1 Identifier Rules

An identifier must be declared or assigned before it is read.

Invalid:

```
OUTPUT total
total <- 0
```

22.2 Scope Rules

Variables declared within a function belong to that function.

Loop variables belong to the containing function or block, depending on the implementation policy.

Function parameters are available throughout the function body.

22.3 Type Compatibility

Operations must use compatible types.

Valid:

```
total <- INTEGER(5) + INTEGER(2)
```

Invalid:

```
total <- INTEGER(5) + STRING("two")
```

22.4 Boolean Conditions

Conditions used by IF and WHILE must evaluate to Boolean values.

22.5 Return Rules

A RETURN statement may only occur inside a function, method, or constructor where explicitly permitted.

Constructors should not return a value.

22.6 Break and Continue Rules

BREAK and CONTINUE may only occur inside loops.

22.7 Collection Rules

List indexes must be integers.

Dictionary keys must be hashable in the generated Python representation.

Collection operations must target compatible collection types.

22.8 Function Call Rules

The number of supplied arguments must match the number of required parameters unless optional

parameters are later added.

22.9 Range Rules

The STEP value must not be zero.

Range loops are end-exclusive.

22.10 Character Rules

A CHAR value must contain exactly one character or one valid escape sequence.

22.11 Member Calls and Dictionary GET Resolution

Property access object.property and member invocation object.method(...) are distinct constructs.

A member call is resolved according to the receiver type or other static evidence.

When the receiver is dictionary-backed and the method identifier is GET, the call resolves to the dictionary intrinsic.

The dictionary intrinsic GET accepts exactly one or two arguments.

When the receiver is not dictionary-backed, GET remains an ordinary user-defined method identifier.

An unknown receiver type must not be assumed to be a dictionary merely because the method is named GET.

NONE is a literal value and is exempt from identifier declaration-before-use checks.

22.12 Nested/Local Function Rules

A function declared inside another function belongs to the lexical scope of the enclosing function and must not be promoted to top-level or module scope.

A nested function may call itself recursively by name and may read values already visible in an enclosing lexical scope.

Assignment to a simple identifier inside a nested function follows that nested function local-binding rule. v1.2 introduces no NONLOCAL-style rebinding construct.

Function declarations are not first-class values in v1.2. Callback parameters, function references, anonymous functions, callable-value storage, and closure escape semantics remain outside the dialect.

23. Normalisation Rules

The evaluation framework may perform conservative normalisation before strict parsing.

Permitted transformations include:

Keyword case normalisation

Unicode arrow ← -> <-

Removal of Markdown code fences

Line-ending normalisation

Trailing whitespace removal

Approved spelling correction

Approved synonym mapping

v1.1 compatibility normalisation may also include:

Legacy dictionary~GET(...) -> dictionary.GET(...)

Case normalisation of None / none -> NONE when used as a standalone null literal

These transformations must obey the same logging and semantic-preservation requirements as all other normalisation.

v1.2 compatibility normalisation may also preserve a source-supplied nested function relationship when lexical nesting is structurally recoverable.

It must not hoist a nested helper, manufacture a nested helper, infer callback semantics, or introduce nonlocal-style rebinding that the source did not provide.

Examples:

while x < 5 do

may become:

```
WHILE x < 5 DO
```

x ← 5

may become:

```
x <- 5
```

The framework must log every transformation.

It must not:

invent missing variables;

insert missing algorithmic operations;

change conditions;

add missing branches;

alter arithmetic operators;

repair semantic logic silently.

24. Guided Link Rules

Guided Links are documentation annotations and do not alter program execution.

Recommended syntax:

```
GUIDED LINK("typed lists")
```

Formal rule:

```
guided_link ::=
```

"GUIDED LINK" ,

"(" ,

quoted_string ,

")" ,

```
newline ;
```

Guided Links may appear between statements and should be:

- stored as metadata;
- excluded from AST execution nodes;
- excluded from generated Python;
- optionally shown in teaching-oriented visualisations.

25. Complete Example

```
FUNCTION find_largest ACCEPTS(numbers AS LIST)
IF LENGTH OF numbers == 0 THEN
RAISE STRING("The list cannot be empty")
ENDIF

largest <- numbers[0]

FOR EACH number IN numbers DO
IF number > largest THEN
largest <- number
ENDIF
ENDFOR

OUTPUT STRING("The largest value is {largest}")
RETURN largest
ENDFUNCTION
```

26. Expected AST Outline

ProgramNode

FunctionNode: find_largest
└──

ParameterNode: numbers AS LIST
├──

Body
└──

IfNode
├──

Condition
│
├──

EqualNode
│
│ └──

LengthNode
│
│
├──

IdentifierNode: numbers
│
│
│
└──

IntegerLiteralNode: 0
│
│
└──

ThenBody
│
└──

RaiseNode

└──

StringLiteralNode
└──

AssignmentNode
├──

IdentifierNode: largest
├──

IndexAccessNode
└──

IdentifierNode: numbers
├──

IntegerLiteralNode: 0
└──

ForEachNode
├──

Iterator: number
│ ├──

Collection: numbers
│ ├──

Body
│ └──

IfNode
└──

GreaterThanNode
├──

AssignmentNode
└──

OutputNode
├──

ReturnNode
└──

27. Versioning Rule

This specification is frozen as:

Cambridge-Style Pseudocode - AJ_24038155 Dialect v1.2

v1.2 supersedes the v1.1 specification by integrating the approved nested/local-function language change while retaining all v1.0 and v1.1 rules:

NONE as a native null literal;

general member-method invocation using object.method(...);

canonical dictionary retrieval using dictionary.GET(...);
backwards-compatible acceptance of the v1.0 dictionary~GET(...) form;

nested/local FUNCTION declarations inside function bodies with lexical scope preservation;

callbacks and first-class function values remain outside v1.2.

Any later syntax addition, removal, or semantic alteration must create a new version, for example:

AJ_24038155 Dialect v1.3

AJ_24038155 Dialect v2.0

AI outputs should always be evaluated against the exact grammar version recorded at generation and evaluation time.